

# When Simple Wins

## Lightweight CNN Encoders for Resource-Constrained Nucleus Segmentation

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With tens of annotated images and one GPU, a plain **VGG U-Net** matches ResNet, beats a pretrained **Swin Transformer** on both benchmarks, and runs faster in less memory.

**+3.0**

Dice over pretrained Swin on PanNuke (0.854 vs 0.824)

**0.794**

MoNuSeg Dice from scratch, at MedT's published level (0.796)

**2.6×**

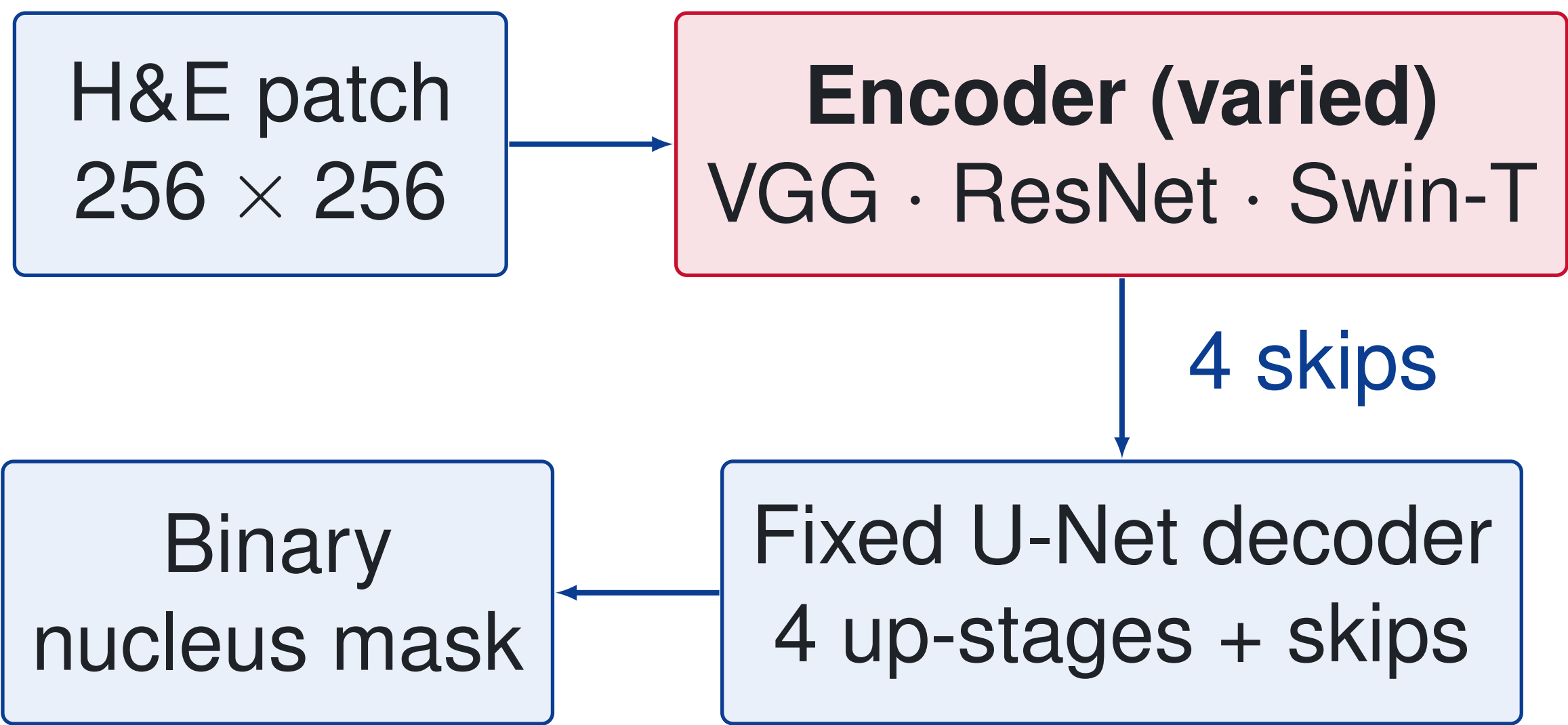
faster than Swin-T at 2.3× less memory

### Motivation

Pathology foundation models (UNI 307M, Virchow2 632M parameters) ease the labelled-data shortage, but they are hard to fine-tune, questioned under cross-site shift, and transfer poorly beyond H&E. Edge microscopy, veterinary and rare-disease labs often have **only tens of annotated slides and a single GPU**.

**Question:** among lightweight, from-scratch-trainable encoders, which is best in the low-data regime?

### Controlled design: only the encoder changes



Decoder, data pipeline, augmentation, optimiser and evaluation are shared; every encoder returns four feature maps through one interface. Results are **5 training seeds** on one V100, with paired 95% CIs.

| Encoder                     | First skip resolution       | Params |
|-----------------------------|-----------------------------|--------|
| VGG (plain 3 × 3 stacks)    | 256 <sup>2</sup>            | 34.0M  |
| ResNet (residual blocks)    | 256 <sup>2</sup>            | 37.5M  |
| Swin-T (scratch / ImageNet) | 64 <sup>2</sup> → upsampled | 86.8M  |

### Datasets

**PanNuke:** 19 tissue types, 256 × 256 patches; train fold1 (2,656 patches), validate fold2, test fold3.

**MoNuSeg:** 37 training slides (1000 × 1000, 7 organs); 14 test slides including organs never seen in training. Tiled into 256 × 256 patches, split by slide.

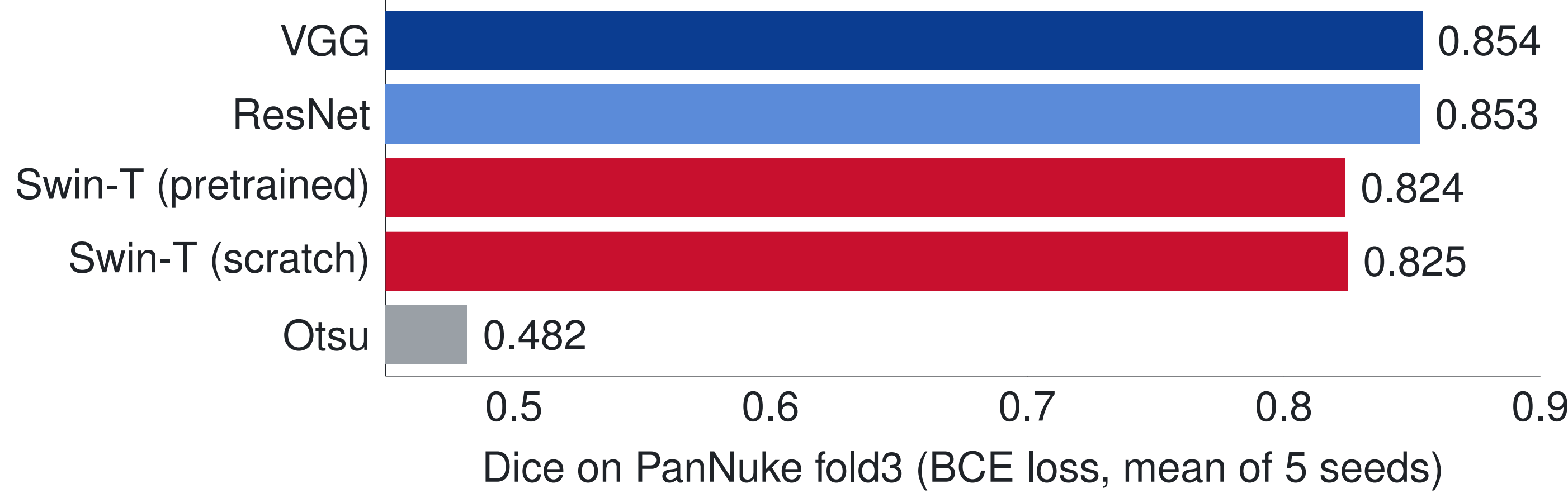
### Why should CNNs win here?

- Skip resolution.** Nucleus boundaries are a few pixels wide. VGG keeps full resolution at its first skip; Swin's 4 × 4 tokens downsample 4× at once.
- Inductive bias vs data.** Attention must learn locality that convolution builds in, and a few thousand patches isn't enough data for that.
- Input scale.** On 256<sup>2</sup> patches, VGG's receptive field already spans most of the input by stage 4.

### Limitations

5 seeds on one fixed split, so split variance isn't bounded. No learning-rate sweep, and the LR schedule differs between CNN and Swin. Only three encoders; at 34M parameters VGG isn't a truly compact model. Binary masks only, not instance separation. H&E only.

### PanNuke: VGG ≈ ResNet > Swin



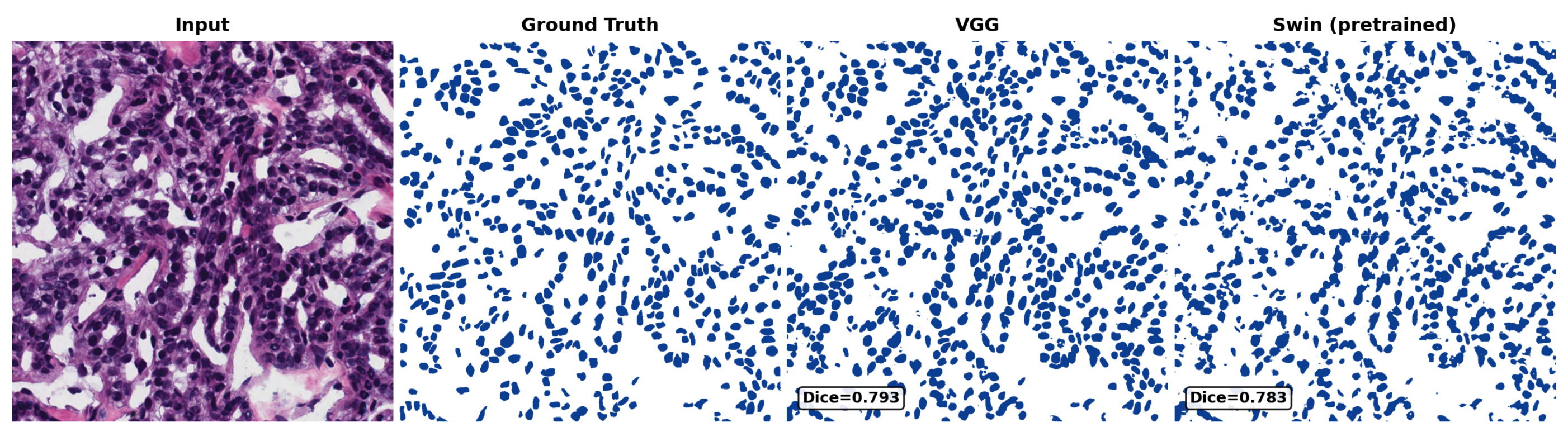
- VGG vs ResNet: 0.002 Dice (CI [0.001, 0.002]), below the 0.005 we treat as meaningful, so **we do not rank them**. Residual shortcuts don't help at four stages.
- VGG vs pretrained Swin: 0.030 **Dice** (CI [0.029, 0.031]).
- ImageNet pretraining moves Swin by only 0.001, so the gap doesn't come from the pretrained weights.
- BCE vs Dice loss: 0.854 vs 0.856, which is immaterial.

### MoNuSeg: 37 training images

| Model                                  | Dice                | IoU                 | Params |
|----------------------------------------|---------------------|---------------------|--------|
| MedT (published, 30 imgs) <sup>†</sup> | 0.796               | 0.662               | 1.4M   |
| Swin-T pretrained (ours)               | 0.758 (.007)        | 0.613 (.008)        | 86.8M  |
| <b>VGG (ours, scratch)</b>             | <b>0.794</b> (.011) | <b>0.660</b> (.015) | 34.0M  |

VGG beats pretrained Swin by **0.036 Dice** (CI [0.021, 0.052]), a gap similar to PanNuke's.

<sup>†</sup>A cross-paper anchor rather than a controlled comparison: MedT used 30 training images and 512<sup>2</sup> input. We read it only as VGG *reaching* MedT's level.



A dense test region, chosen for how dense it is rather than for the size of the gap. Swin merges neighbouring nuclei across narrow gaps that VGG keeps separate.

### Inference cost (256<sup>2</sup>, batch 1, fp32, V100)

| Model            | Params | Latency       | Peak mem.     |
|------------------|--------|---------------|---------------|
| <b>VGG U-Net</b> | 34.0M  | <b>7.1 ms</b> | <b>265 MB</b> |
| ResNet U-Net     | 37.5M  | 9.6 ms        | 277 MB        |
| Swin-T U-Net     | 86.8M  | 18.4 ms       | 619 MB        |

**VGG is better than Swin-T on accuracy, latency and memory.** Every model we trained fits in 8 GB (peak training memory: 3.6 GiB for VGG, 5.4 GiB for Swin).

### Next steps

A resolution-matched Swin to separate tokenisation from encoder family; H&E-matched pre-training (e.g. UNI); the data scale at which Swin catches up; compact encoders (MobileNet, ConvNeXt-T); non-H&E stains.

**Code & checkpoints:** [github.com/Bedrockdude10/BinaryCellSegmentation](https://github.com/Bedrockdude10/BinaryCellSegmentation)